BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Linear Search

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is a syntax error?

2. What does FROM name in SQL?
   A) The output columns
   B) The source table
   C) The filter condition
   D) The sort order

3. "Encryption prevents all network packets from being intercepted." Correct this. Explain why.

4. A school replaces working laptops each year. "This has no environmental cost." Correct this. What resources or waste are involved?

Today's objectives

Explain that linear search examines elements one by one. Trace a successful and unsuccessful search.

Key vocabulary - Read, cover, recall

Linear search
Method checking array elements in sequence until a match is found.

Target
Value being searched for.

Comparison
Checking an array value against the target.

Index
Zero-based position in an array.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New Learning - Key ideas

Search each position

Linear search starts at the first element, compares it with the target, and moves through elements in order.

Works with unsorted data

The values do not need to be sorted. Linear search can inspect a list containing any arrangement.

Record comparisons and result

Stop once the first target is found (or record it without overwriting), or report not found after every relevant element is checked.

Use array indices

OCR ERL arrays are zero indexed. The fourth element is at index 3; the number of comparisons and array position are not the same.

I Do - Worked model

Trace a linear search for target 2 in [8,4,11,2,7]. State the checked values and zero-based position where 2 is found. [4 marks]

- Begin at index 0 and check 8; it is not 2 [1].
- At index 1, check 4; still not 2 [1].
- At index 2, check 11; still not 2 [1].
- At index 3, value 2 matches. Report position 3 [1].

We Do - Guided practice

1. Trace linear search for 5 in [3,9,5]. State the comparisons and the index found. [3 marks]

2. Search for 7 in [8,6,4]. Explain the outcome and number of comparisons. [3 marks]

3. Search [2,2,5] for 2 using linear search. Explain the first match position and why. [2 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that finds "PC" in a register-name array.

```
array names = ["MAR","MDR","PC"]
for i = 0 to 2
if ... then
print(i)
endif
next i
```

You Do - Independent exam questions

Q2. State one advantage of linear search on an unsorted array. [2 marks]

Q3. A list is [6,3,8,1]. Trace linear search for 1, stating each comparison and index. [4 marks]

Q4. A shop has an unsorted array [5,9,2,7,3]. Explain the main steps of linear search for 7, compare successful and unsuccessful outcomes and describe how an index is reported. [6 marks]

Exit Ticket - True or False?

Linear search requires an array to be sorted before it can be used.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A syntax error breaks the language's rules. The affected code cannot be translated or run correctly.

2. B - The source table

3. Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key.

4. Making devices uses energy and raw materials. Disposal adds electronic waste. Consider who benefits and who is harmed.

I Do - Explain the model

Begin at index 0 and check 8; it is not 2 [1].

At index 1, check 4; still not 2 [1].

At index 2, check 11; still not 2 [1].

At index 3, value 2 matches. Report position 3 [1].

We Do - Model answers

1. Check 3 then 9 [1]; check 5 [1]; index found = 2 [1].

2. Compare 8, 6 and 4 [1]; no element equals 7 [1]; report not found after 3 comparisons [1].

3. The first element is 2 [1], so the first matching index is 0 [1].

Code First - Answer

```
array names = ["MAR","MDR","PC"]
for i = 0 to 2
if names[i] == "PC" then
print(i)
endif
next i
```

Expected output: 2

You Do - Self-mark

Q2: It does not require sorted data [1], so it can search the current unsorted array without sorting it first [1].

Q3: Compare 6 [1], then 3 [1], then 8 [1], then 1 at zero-based index 3 [1].

Q4: Start at index 0 [1], compare 5 then 9 then 2 [1]; at index 3 value 7 matches [1]. Return index 3 [1]. If all entries were checked without a match [1], report not found, e.g. -1 [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.